

Question Number	Answer	Additional Guidance	Mark
3(a)	There is no (significant) difference between the (mean) time before extension in mud and sand (habitat)	Accept equivalent correct answers Eg time to re-emerge/ different substrates	(1)

Question Number	Answer	Additional Guidance	Mark																										
3(b)(i)	- suitable table with correct raw data (1) - with headings with units (1) - correct means (1)	MP1 do not accept an extra column Time before extension /s <table><tr><td>Mud (habitat)</td><td>Sand (habitat)</td></tr><tr><td>14.8</td><td>14.0</td></tr><tr><td>8.2</td><td>7.4</td></tr><tr><td>9.9</td><td>9.2</td></tr><tr><td>11.6</td><td>12.6</td></tr><tr><td>12.0</td><td>10.7</td></tr><tr><td>14.2</td><td>7.8</td></tr><tr><td>10.7</td><td>8.0</td></tr><tr><td>11.2</td><td>11.1</td></tr><tr><td>8.9</td><td>10.7</td></tr><tr><td>12.1</td><td>10.8</td></tr><tr><td>12.9</td><td>11.2</td></tr><tr><td>Mean 11.5</td><td>Mean 10.3</td></tr></table> Means must be to the same number of decimal places	Mud (habitat)	Sand (habitat)	14.8	14.0	8.2	7.4	9.9	9.2	11.6	12.6	12.0	10.7	14.2	7.8	10.7	8.0	11.2	11.1	8.9	10.7	12.1	10.8	12.9	11.2	Mean 11.5	Mean 10.3	(3)
			Mud (habitat)	Sand (habitat)																									
			14.8	14.0																									
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3(b)(ii)	- axes labelled including units, and linear scale starting at zero without a break (1) - data plotted correctly in bar graph format (1) - range bars correctly plotted (1)	Mean time (for extension) /s mud sand Ignore line if drawn Mean 11.5 10.3 Range mud 8.2 -14.8 sand 7.4 - 14	(3)

Question Number	Answer	Additional Guidance	Mark
3(c)(i)	- correct numerator (1) - correct denominator (1) - correct answer (1)	Allow ECF if incorrect mean values used for MP2 and 3 1.2/1.182 0.763 /0.873 1.37 /1.35 Correct answer gains 3 marks Correct substitution into formula gains one mark instead of mps 1 and 2	(3)

Question Number	Answer	Additional Guidance	Mark
3(c)(ii)	- the critical value is 2.09 (1) - calculated value is less than the critical value, therefore accept the null hypothesis (1) - there is no (significant) difference between the extension time for mud and sand animals (1) - comment on variability of data (1)	Accept ECF for calculated values greater than 2.09 Followed by logical statements Accept there is a difference if they have rejected null hypothesis Eg range bars overlap	(3)

Question Number	Answer	Additional Guidance	Mark
3(d)	An answer that includes two of the following: - opsin changes when stimulated by {light/photon} / changes retinal from cis to trans / a different isomer (1) - causes an {action potential/impulse} (in sensory) nerve (1) - causing a reflex response (1)	Ignore refs to rhodopsin Accept descriptions leading to hyperpolarisation Accept muscle contraction/reflex action	(2)

(Total for question 3 = 16 marks)

